

Module 13 : python fundamentals

Introduction to Python and its Features.

1.what is python ?

- python is a high-level , enterprited , object-oriented and multipurpose programming language.

2. python Features.

- High-Level Language
 - requiried interprited and compiler.
- Interpreted Language
 - line by line by the execute the code.
- Object-Oriented and Functional
 - Supports object-oriented programming (OOP) with classes and objects.
- Portable
 - Write once, run anywhere (as long as Python is installed).
 - Works on Windows, Mac, Linux, etc.
- Dynamically Typed

- No need to declare variable types.

History and evolution of Python.

➤ **Who created Python?**

- Guido van Rossum, a Dutch programmer, started working on Python in the late 1980s.
- 1980 : thinking.
- 1989 : Python Begins.
- 20-feb-1991 : First version official release.
- 02-cot-2023 : latest

Advantages of Using Python Over Other Programming Languages.

- Python has several advantages over other programming languages, making it one of the most popular choices for developers across various domains. Here are the key advantages:

1. Readability and Simplicity.

- Python's syntax is clear and resembles natural language, making code easier to read, write, and maintain - especially for beginners and teams.

2. Wide Range of Libraries and Frameworks.

- Python has an extensive standard library and many third-party modules for various domains, including:
- Web development: Django, Flask.
- These save development time by providing ready-to-use tools.

3. Versatility and Cross-Platform Compatibility.

- Python is a general-purpose language used in:
 - * Web and software development
 - * Data analysis and machine learning
 - * Scientific computing
 - * Automation
 - * Game development
- It runs on major platforms (Windows, macOS, Linux) without needing to change the source code.

4. Interpreted Language

- Python is an interpreted language, meaning:
- Interpreted means code is executed line-by-line .
- There's no need to compile the code, which speeds up the development cycle.



Installing Python and Setting Up the Development Environment (Anaconda, PyCharm, and VS Code).

- Before writing Python code, it's important to install Python and choose a development environment that suits your needs.

Below are the steps and options:

1. install python on your local machine

- download python
- go to python.org
- download the latest version for your operating system (windows , macos or linux)
- run the installer
- verify installation
- open a terminal/command prompt
 - type `python --version` or `python3 --version` and press enter
 - you should see the installed python version

1. Using Anaconda

- Anaconda is a distribution that comes pre-installed with Python and many popular data science libraries (NumPy, pandas, Matplotlib, Jupyter Notebook).
- Steps
 1. Download-Anaconda-from
[<https://www.anaconda.com>]
 2. Follow the installer instructions.
 3. Open Anaconda Navigator or use Anaconda Prompt.

2. Using PyCharm

- PyCharm is a full-featured IDE developed by JetBrains, ideal for larger projects.
- Steps:
 1. Download PyCharm
<https://www.jetbrains.com/pycharm>
 2. Install and run it.
 3. Create a new project and set the Python interpreter.

3. Using Visual Studio Code

- VS Code is a lightweight and powerful code editor with Python support via extensions.
- Steps
 1. Download and install VS Code from
<https://code.visualstudio.com>
 2. Open VS Code and install the Python extension from Microsoft.
 3. Select or install a Python interpreter.



Writing and executing your first Python program.

➤ Using a Simple Text Editor and Command Line.

➤ Steps

Open text editor (vs code)

Print("hello world")

Save the file as hello.py.

➤ To run

Open terminal.

Run the program.

➤ Conclusion:

This confirms that Python is installed and working correctly on your system